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Robot Cable Assembly with Machine Vision and Force Estimation

Capstone Project Report

Electromechanical Engineering
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Abstract

Manual wire terminal assembly remains a significant bottleneck in modern manufacturing, particularly in high-mix production environments where multiple terminal types must be handled and inserted into predefined sockets. These processes are often time-consuming, prone to human error, and difficult to scale using conventional automation, which typically relies on dedicated tooling with limited flexibility. The objective of this capstone project is to design and validate an autonomous robotic system that improves productivity and flexibility in wire terminal assembly by enabling the handling and insertion of different terminal types without manual intervention.

The proposed system integrates machine vision for wire terminal classification with a robotic manipulator equipped with a mechanically compliant gripper. The gripper is specifically designed with an embedded spring-based compliance mechanism that allows indirect estimation of insertion force based on spring deflection, providing passive force control during the insertion process. Known socket locations are used to simplify motion planning and ensure repeatable insertion. The mechanical design of the gripper, including spring selection and force modeling, is analyzed to ensure reliable insertion while preventing damage to terminals and sockets.

The system is evaluated through experimental testing based on insertion success rate, cycle time, and force consistency. The results demonstrate the feasibility of using a compliant, mechanically simple gripper to enhance flexibility and efficiency in automated wire terminal assembly, addressing key limitations of existing manual and semi-automated processes.

Motivation and Purpose

The increasing demand for customized products and shorter production cycles has intensified the challenges associated with electrical wire assembly in manufacturing environments. Despite advances in automation, wire terminal assembly remains largely manual, particularly in high-mix production scenarios where multiple terminal types must be handled and inserted into predefined sockets. These tasks are repetitive, time-consuming, and susceptible to human error, leading to inconsistent quality, extended cycle times, and limited scalability. Conventional automated solutions often rely on dedicated tooling designed for a single terminal type, resulting in long setup times and reduced flexibility when product variants change.

The motivation for this project arises from the need to bridge the gap between manual assembly and rigid automation by developing a flexible, mechanically robust solution capable of handling product variability without sacrificing productivity. From an industrial perspective, improving cycle time and reducing assembly errors directly impact manufacturing efficiency and cost. From an academic standpoint, the project provides an opportunity to apply and integrate core mechanical engineering principles, such as compliant mechanism design, force modeling, and tolerance analysis, within an automated robotic system.

The primary purpose of this capstone project is to design, implement, and evaluate an autonomous robotic wire terminal assembly system that enhances productivity in high-mix production environments. By incorporating machine vision for terminal identification and a mechanically compliant gripper with spring-based force sensing, the system aims to achieve reliable insertion of different terminal types while minimizing setup complexity. The project ultimately seeks to demonstrate that mechanical compliance and intelligent system integration can significantly improve flexibility and efficiency in automated wire assembly processes.

Methods

1. System Architecture and Overall Approach

The proposed system is designed to autonomously perform wire terminal assembly by integrating robotic manipulation, machine vision, and a mechanically compliant end-effector. The methodology focuses on improving productivity and flexibility in high-mix production environments by reducing reliance on dedicated tooling and precise force control hardware. The system consists of four main subsystems: a robotic manipulator, a vision-based terminal classification module, a compliant gripper with spring-based displacement sensing, and a fixed socket fixture with predefined terminal locations.

Wire terminals are presented in a dedicated fixture with predefined position and orientation. This deterministic presentation allows the vision system to focus exclusively on terminal type classification rather than pose estimation, thereby simplifying system integration and increasing repeatability. Once a terminal type is identified, the robot retrieves the corresponding socket location from a predefined database and executes an autonomous pick-and-insert sequence.

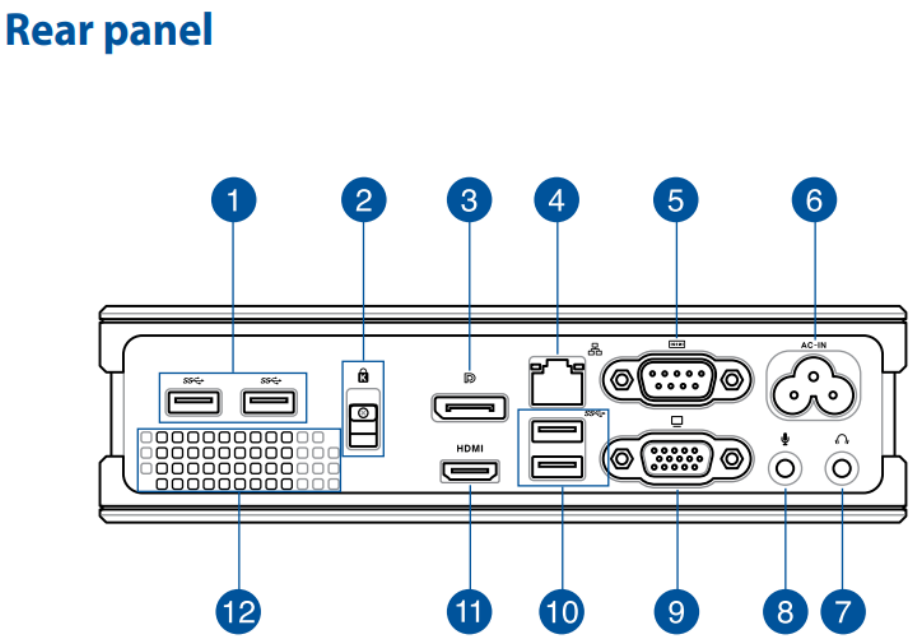

Figure 1: ASUS VC65 Connectors Locations

1	USB 3.0 port	7	Headphone jack
2	Top cover lock latch + Kensington® security slot	8	Microphone jack
3	DisplayPort	9	VGA port
4	LAN port	10	USB 3.0 port
5	Serial (COM) connector	11	HDMI port
6	AC-IN Power (AC) input port		

Table 1: Device ports and connectors

2. Wire Terminal Classification Using Machine Vision

Machine vision is employed solely to classify wire terminal types prior to manipulation. Classification is based on two-dimensional geometric features such as contour shape, terminal width, and characteristic openings, which are sufficiently distinct among the selected terminal types. A fixed camera captures images of the terminals, and image processing techniques are applied to identify the terminal category.

By constraining the pickup position and orientation mechanically, the vision system is not required to perform pose estimation or three-dimensional reconstruction. This approach reduces computational complexity and increases robustness while maintaining the flexibility required to handle multiple terminal types without mechanical reconfiguration.

3. Compliant Gripper Design with Spring-Based Displacement Sensing

The mechanical design of the gripper represents the core contribution of this project. The gripper is designed to securely grasp different wire terminal geometries while providing axial compliance during the insertion process. Compliance is achieved through the integration of a linear compression spring aligned with the insertion axis.

In addition to providing passive compliance, the spring functions as a displacement-based sensing element for misalignment detection. Under normal insertion conditions, the terminal is expected to enter the socket with minimal resistance, resulting in limited spring compression. In the presence of misalignment between the terminal and the socket, increased resistance causes the spring to compress beyond a predefined displacement threshold.

The relationship between insertion force and spring compression is described by Hooke's law:

$$F = k \cdot \Delta x$$

where F is the axial force, k is the spring constant, and Δx is the measured spring compression. The spring constant and maximum allowable compression are selected based on expected insertion forces and mechanical constraints. Mechanical hard stops are incorporated to prevent excessive compression and protect both the gripper mechanism and the socket interface.

Closed-loop Force Control from Desired Force to Applied Force

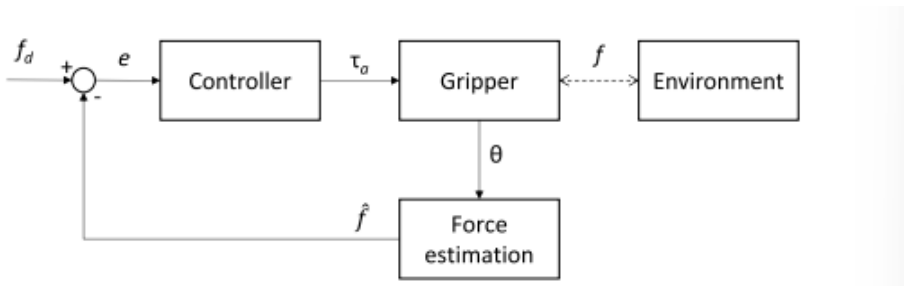

Figure 2: Simple Schematic Diagram Reference

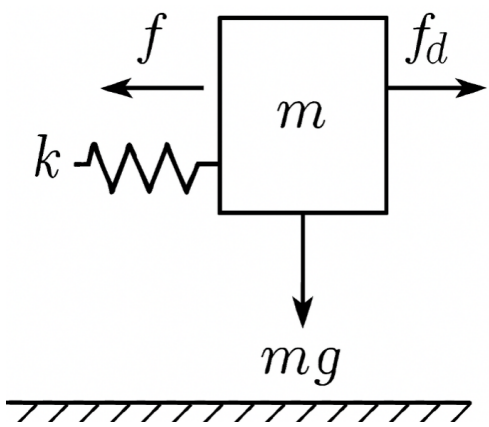

Figure 3: Free Body Diagram of Forces for the Gripper System

Block Relationships with Measurement

- **Error:**

$$E(s) = F_d(s) - \hat{F}(s)$$

- **Controller:**

$$U(s) = C(s)\,E(s)$$

- **Plant (linear axis + spring):**

$$G(s) = \frac{F(s)}{U(s)} = \frac{k}{ms^2 + bs + k}$$

- **Force estimation (measurement path):**

$$\hat{F}(s) = H(s)\,F(s)$$

- **Special cases:**
  - Direct force sensor: $H(s) = 1$
  - Estimate via displacement: $\hat{F}(s) = k\,X(s) = F(s) \Rightarrow H(s) = 1$
  - Filtered estimate: $H(s)$ could be a low-pass

$$H(s) = \frac{1}{1 + s/\omega_f}$$

Closed-loop Transfer Function

Starting from:

$$F(s) = G(s)\,U(s) = G(s)\,C(s)\,[F_d(s) - H(s)\,F(s)]$$

$$F(s)\,[1 + G(s)\,C(s)\,H(s)] = G(s)\,C(s)\,F_d(s)$$

Therefore, the closed-loop transfer function from desired force to applied force is:

$$\frac{F(s)}{F_d(s)} = \frac{\dfrac{k\,C(s)}{ms^2 + bs + k}}{1 + \dfrac{k\,C(s)}{ms^2 + bs + k}\,H(s)}$$

Sensitivity to Disturbances

If a disturbance force $F_{\text{ext}}(s)$ adds at the plant output (environment reaction), the output becomes:

$$F(s) = \frac{G(s)\,C(s)}{1 + G(s)\,C(s)\,H(s)}\,F_d(s) \;+\; \frac{1}{1 + G(s)\,C(s)\,H(s)}\,F_{\text{ext}}(s)$$

Key functions:

$$T(s) = \frac{G(s)\,C(s)}{1 + G(s)\,C(s)\,H(s} \quad \text{(Complementary sensitivity: command tracking)}$$

$$S(s) = \frac{1}{1 + G(s)\,C(s)\,H(s} \quad \text{(Sensitivity: disturbance/noise rejection)}$$

Controller Choice and Practical Notes

- **PID controller:**
$$C(s) = K_p + \frac{K_i}{s} + K_d s$$
- K_p: raises loop gain for stiffness and tracking.
- K_i: eliminates steady-state error to a constant f_d.
- K_d: adds damping to tame oscillations from the mass–spring.
- **Measurement dynamics:** If $H(s) \neq 1$ (filtering), ensure the filter cutoff is well above the loop bandwidth to avoid excess phase lag.
- **Tuning target:** Achieve adequate phase margin (robustness) while keeping the closed-loop bandwidth below resonance peaks from $G(s)$. Start with PI (no K_d), then add small K_d if overshoot persists.

The overall design emphasizes mechanical simplicity, repeatability, and robustness.

4. Insertion Strategy and Misalignment Correction

The insertion process follows a structured motion sequence consisting of approach, insertion, verification, and release phases. During the insertion phase, the robot advances the terminal toward the socket at a controlled speed. Spring compression is continuously monitored to detect abnormal resistance indicative of misalignment.

Control Architecture

Low Level: Force Control

- **Input:** f_d (desired force)
- **Output:** f (applied force on the spring/cable)
- **Feedback:** $\hat{f}$ (estimated or measured force)

- **Function:** Ensures the gripper presses up to the commanded force.

Dynamic model:

$$G(s) = \frac{F(s)}{U(s)} = \frac{k}{ms^2 + bs + k}$$

At this level, the system does not decide whether insertion succeeded or failed; it only regulates the force.

High Level: Insertion Supervisor

- **Inputs:** Pin1, Pin2, Pin3 (insertion result signals)
- **Outputs:**
  - Target position $(x_{\text{cmd}}, y_{\text{cmd}})$ for the position controller
  - Command to enable/disable the force loop (set $f_d > 0$ or $f_d = 0$)

Compact logical equations:

$$\text{If Pin1} = 1 \text{ or Pin2} = 1\text{: State} \leftarrow \text{HOLD_FINAL}$$

$$\text{If Pin3} = 1\text{: State} \leftarrow \text{RETRACT}, \quad (x_{\text{cmd}}, y_{\text{cmd}}) \leftarrow P_{i+1}$$

Figure 4: Positions to try (9 Positions)

- HOLD_FINAL: hold position, insertion completed - RETRACT: return to neutral, disable force, select the next test point and retry In case of error, try the positions shown

in Figure 4, which is shaped like a gear shift lever. The (Neutral) position is the original point to which I assign the claw. Then try the left and right positions. If you still cannot

insert it, try the other corners, which would be the gears of a car's gearbox: 1, 2, 3, 4, 5, R. After the 9th attempt, stop trying to insert it, release the cable into a container, and proceed with the next cable.

Connection Between Levels

- The supervisor decides which point to test and when to apply force.
- The position controller moves the gripper to the chosen point.
- The force controller regulates the insertion force.
- The pins inform the supervisor whether the insertion was successful or failed.

If the measured spring displacement exceeds a predefined threshold, the system interprets this condition as a misalignment event. In response, the robot retracts the terminal and executes a corrective alignment motion before attempting reinsertion. This threshold-based misalignment detection and correction strategy allows the system to recover from positioning errors without relying on complex force control algorithms or active force sensors.

Mechanical compliance serves as the primary means of error tolerance, while simple control logic enables reliable and repeatable insertion across different terminal types.

5. Experimental Setup and Evaluation Plan

An experimental setup is being developed to evaluate the proposed system, consisting of a fixed socket fixture, a representative set of wire terminal types, and the robotic assembly system. The evaluation focuses on verifying the effectiveness of the compliant gripper and misalignment detection strategy.

The respective claw will have 3 reference positions, thus having 3 pins on the displacement axis with signals depending on the compression of the same, thus giving 3 displacements of the spring which would indicate: correct insertion in average force, correct insertion with excess force or problems, and maximum compression (insertion error).

Performance will be assessed using objective metrics, including insertion success rate, frequency of misalignment detection events, assembly cycle time, and repeatability across multiple trials. Spring displacement data will be used with a closed-loop system to validate threshold selection and distinguish between normal and abnormal insertion conditions.

Similarly, we proceeded to test some simulations in matlab to analyze the behavior of our system.

```
% Closed-loop force control with supervisor logic in MATLAB

% Parameters of the plant
m = 0.5;    % mass [kg]
b = 2;      % damping [Ns/m]
k = 50;     % spring constant [N/m]

% Controller gains (PI example)
Kp = 100;
```

```
Ki = 200;
Kd = 0;

% Plant G(s) = k / (m s^2 + b s + k)
numG = [k];
denG = [m b k];
G = tf(numG, denG);

% Controller C(s) = Kp + Ki/s + Kd*s
C = tf([Kd Kp Ki], [1 0]);

% Closed-loop transfer function T(s) = GC / (1 + GC)
T = feedback(G*C, 1);

% Simulation parameters
t = 0:0.01:2; % time vector
Fd = 10;      % desired force [N]

% Step response of force loop
[y, t] = step(Fd*T, t);

% Plot force response
figure;
plot(t, y, 'b', 'LineWidth', 2); hold on;
yline(Fd, 'r--', 'Desired force f_d');
xlabel('Time [s]');
ylabel('Force [N]');
title('Closed-loop force response');
legend('Applied force f(t)', 'Desired force f_d');
grid on;

%% Supervisor logic simulation
MAX_ATTEMPTS = 9;
attempts = 0;
states = {}; % store state transitions

% Example sequence of pin signals (simulate failures then
   success)
% 0 = no pin, 1 = Pin1 success, 2 = Pin2 success, 3 = Pin3
   failure
pin_sequence = [3 3 3 3 3 3 3 3 3 1]; % 9 fails then success

for i = 1:length(pin_sequence)
    pin = pin_sequence(i);
    if pin == 1 || pin == 2
        states{end+1} = 'HOLD_FINAL';
        break; % success, stop
    elseif pin == 3
        attempts = attempts + 1;
        if attempts >= MAX_ATTEMPTS
            states{end+1} = 'FAIL_STOP';
```

```
            break; % stop after 9 fails
        else
            states{end+1} = sprintf('RETRACT (attempt %d)',
               attempts);
        end
    else
        states{end+1} = 'SELECT_POINT';
    end
end

% Display supervisor timeline
disp('Supervisor state transitions:');
disp(states);

% Plot supervisor timeline
figure;
plot(1:length(states), 1:length(states), 'o-');
xticks(1:length(states));
xticklabels(states);
xlabel('Step');
ylabel('State index');
title('Supervisor state transitions');
grid on;
```

Listing 1: MATLAB closed-loop force control with supervisor logic

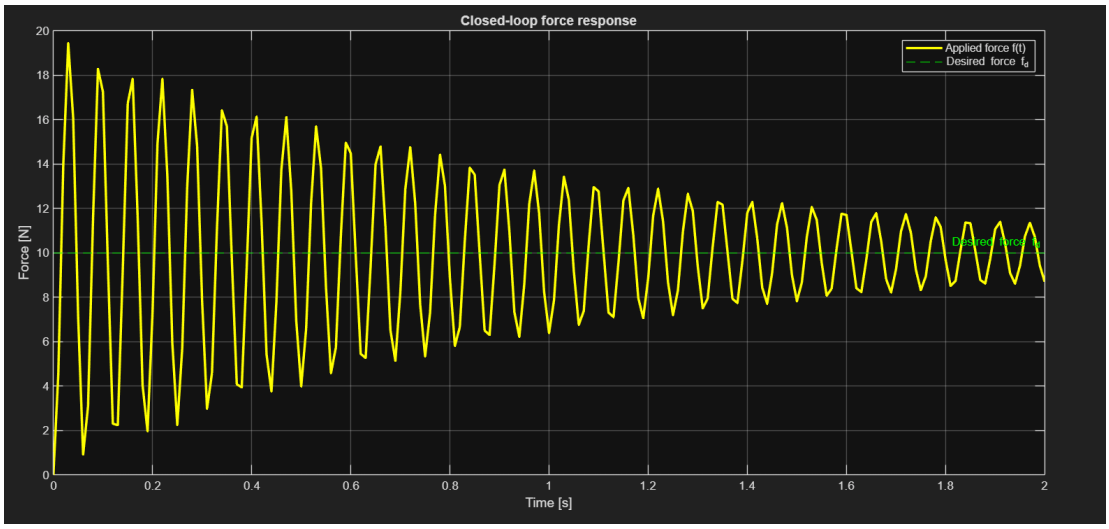

Figure 5: Closed-Loop Force Response (Force Vs. Time)

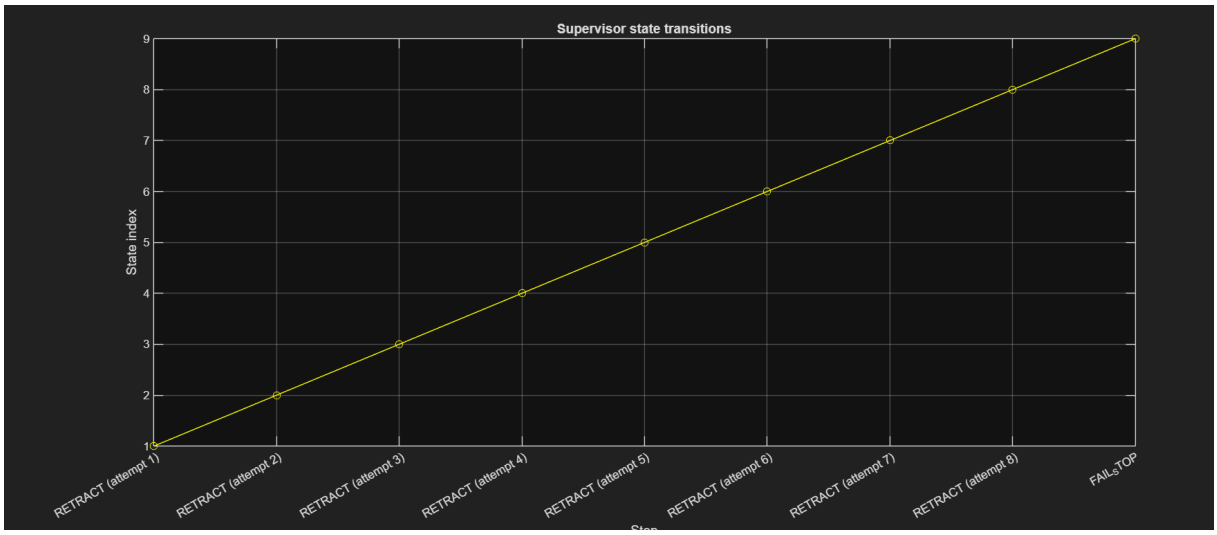

Figure 6: State Transitions

Initial testing is focused on subsystem validation and qualitative performance assessment. As development progresses, quantitative testing will be expanded to enable comparison with manual assembly processes.

6. Development Status

At the current stage of the project, the system architecture and methodological framework have been defined, and mechanical prototyping of the compliant gripper is in progress. Vision-based classification and misalignment detection logic are under development. The methodology presented provides a structured foundation for continued implementation, testing, and refinement leading toward final system validation.

Duty Allocation of the Team Members

The tasks were developed collaboratively as a team, but specific members were responsible for the following areas:

- **Juan**: Robot Programming
- **Saul**: Machine Vision
- **Nicolas**: Gripper Design

Work Log

The project began during the summer break at the end of June. Weekly meetings were conducted to introduce the project objectives, team roles, and overall workflow. During this initial phase, training sessions were carried out to acquire the technical skills required for the successful development of the project. These training activities were organized into three main areas.

First, **ABB RobotStudio** was studied to understand robot motion control, simulation environments, and coordinate definition. This included defining the tool center point (TCP), workspace manipulation, and trajectory planning within simulated industrial scenarios.

Second, **machine vision programming** was explored using Python. The focus was placed on understanding stereo camera operation, feature detection, and distance estimation by comparing corresponding points in dual images. This phase provided the foundation for later visual recognition tasks required by the system.

Third, **training on electronic component management** was conducted. This included learning the basic operation and integration of stepper motors, Raspberry Pi platforms, and AC/DC power supplies, enabling the design of mechanical, electrical, and mechatronic subsystems relevant to the project.

From September to December, the team evaluated possible modifications to the original project, which initially focused on inserting thin wires into terminals. After several group discussions and feasibility analyses, a new approach was proposed: the design of an autonomous cable assembly robot capable of handling USB, HDMI, and other cable types.

To achieve autonomy, **machine vision–based cable terminal recognition** was selected as the core strategy. The objective is to identify the cable type and determine the corresponding insertion terminal to ensure accurate and reliable assembly.

To simulate an industrial assembly line environment, a **rotating cable selector mechanism** was conceptualized. Different cable types are placed at predefined positions on a circular plate, which rotates continuously, allowing the robot to detect and classify the cable currently in operation. To achieve the required rotational speed and torque, a gearbox mechanism using gears and belts driven by a stepper motor was designed. All mechanical components were modeled with the intention of manufacturing them using 3D printing techniques.

The plan for the next semester is to implement the proposed mechanical, vision, and control systems, followed by experimental evaluation to assess feasibility, performance, and potential design improvements.

Budget and Reimbursement Report

The project budget supports the design and development of an autonomous cable assembly robot. The primary budget allocation covers electronic components, mechanical parts, prototyping materials, and vision system hardware required for system implementation and testing.

During the initial phase of the project, no major expenses were incurred, as training activities and preliminary simulations were conducted using laboratory equipment and licensed software provided by the institution. Minor expenses related to basic electronic components and prototyping materials were covered by the lab.

For the next semester, additional expenses are anticipated for mechanical fabrication and system integration. These include 3D printing materials, gearbox components, stepper motor accessories, machine vision components, and supporting electronic hardware. These costs are necessary to enable full system implementation and experimental evaluation.

The elements requested for future advancement in the project are listed in the following table:

Element	**Estimated Price (NTD)**
X2 units of UHD cables	254
X2 units of USB cables	180
X2 units of HDMI cables	160
X1 unit TPU 1.75mm 1KG 3D printing filament	520
X1 unit PLA PETG ABS PA 1.75mm 1KG	600
X1 unit RPG Model Resin 1Kg	1300
X1 unit 8K Model Resin 1Kg	1100
X1 unit Force sensor	130
X10 units Tact switch	10
X1 Spring kit	320
X2 Industrial Plug	332
X2 Industrial Socket	424
Current total estimated cost	**5330**

Table 2: Current Project Budget Allocation

Difficulties Encountered

During the current phase of the project, the main difficulties encountered were related to time constraints and project scope definition. Although initial training and conceptual exploration were conducted, limited time availability during the academic semester restricted the depth of practical implementation.

Another challenge involved defining the project objectives. The transition from the original thin-wire insertion task to an autonomous cable assembly system required additional discussions and feasibility analysis, which delayed the start of detailed technical implementation.

Additionally, the complexity of the proposed system combining mechanical design, electronics, machine vision, and robotic control necessitated careful planning and staged development. As a result, priority was given to conceptual design and system architecture rather than immediate implementation.

These challenges have informed the planning of the next project phase, which will focus on gradual implementation, tool-specific experimentation, and system integration based on the established design framework.

Photos

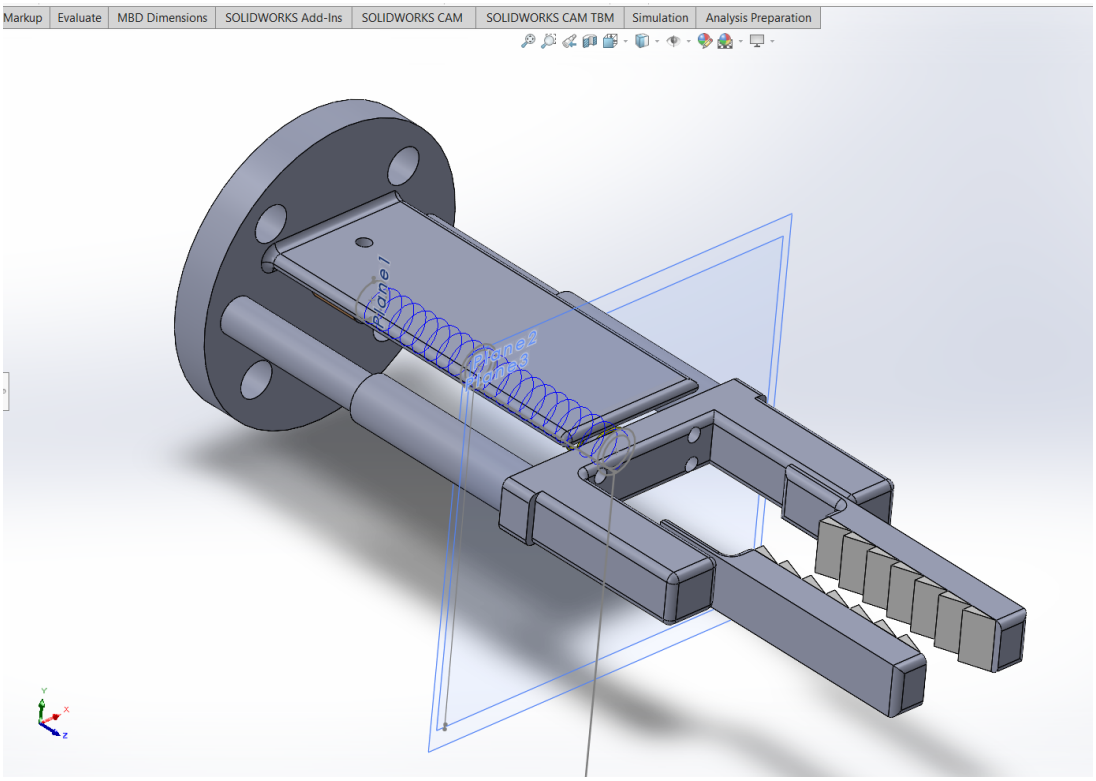

Figure 7: Gripper CAD

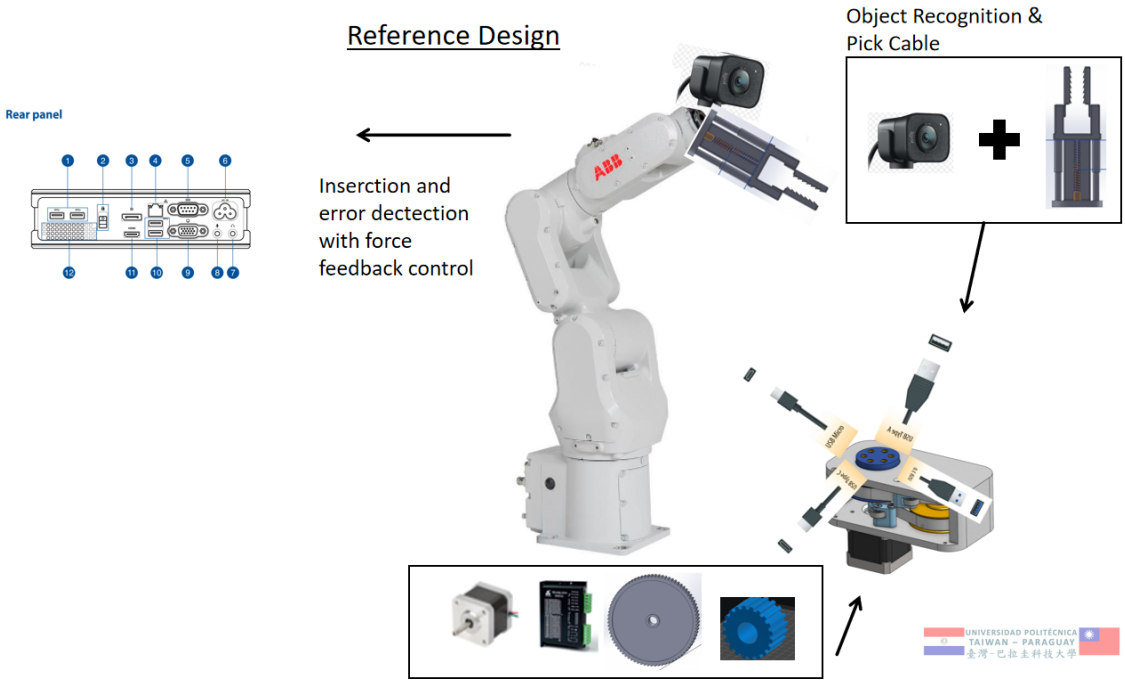

Figure 8: Design Reference

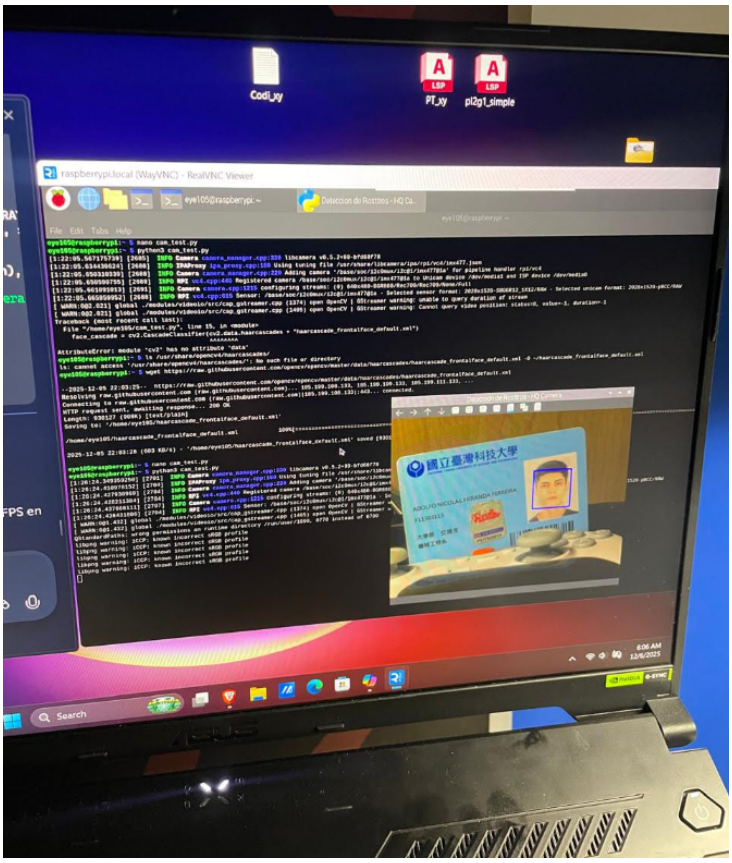

Figure 9: Camera Test

Figure 10: Design Gripper Inspiration

Feedback from the Team Members

Saul: During this process of the project, I could develop many new skills related to mechanical, electrical, electronic and programming areas. I am really thankful to the professor and his Lab for letting us access to many top innovative technologies and let us flow with our imagination. I really liked to work with my team and hope to continue with the project implementation and end a successful process.

Juan: Being responsible for the robot programming in this project provided a valuable learning experience that significantly strengthened my understanding of robotic systems and automation. Through the development of motion sequences, error handling strategies, and integration with peripheral components, I gained practical insight into the challenges of translating theoretical concepts into reliable robotic behavior. The project also highlighted the importance of structured programming and systematic testing when working with physical systems. I am grateful for the guidance and resources provided by the course instructor, which made it possible to explore these topics in depth and apply them in a realistic engineering context. Overall, this project contributed meaningfully to my technical skills and reinforced my interest in robotic automation and mechatronic system design.

Nico: I am very grateful to the professor for providing us with all the resources at his disposal, even more than we need. I feel like I'm in a space where I can develop my creativity with good guidance from both the professor and the teaching assistants. This project is making me feel like an engineer because I'm applying many theories I learned in previous semesters to real-life cases. I could learn a lot from mechanical, electrical, electronic circuits and its application and be in touch with a complete lab.

Conclusion and Review

This project presents the conceptual design and initial development of an autonomous robotic system for wire and cable terminal assembly, aimed at improving productivity and flexibility in high-mix manufacturing environments. The proposed approach addresses key limitations of conventional manual and semi-automated assembly processes by integrating robotic manipulation, machine vision–based terminal classification, and a mechanically compliant gripper designed for reliable insertion without complex force control hardware.

At the current stage, the system architecture and methodological framework have been clearly defined. The use of machine vision for terminal type classification, combined with deterministic mechanical positioning, simplifies perception requirements while maintaining flexibility across multiple terminal variants. The compliant gripper design, incorporating a spring-based displacement mechanism, represents a mechanically simple yet effective solution for passive force control and misalignment detection during insertion. Theoretical force modeling based on Hooke's law provides a solid foundation for selecting appropriate compliance parameters and protecting both terminals and sockets from damage.

Although full system implementation and quantitative experimental validation are still in progress, the design decisions and preliminary development demonstrate the technical feasibility of the proposed concept. The challenges encountered—particularly those related to time constraints, project scope refinement, and system complexity—have guided the prioritization of architectural design and subsystem development during this phase. These experiences have contributed to a more realistic and structured plan for the next stage of the project.

Future work will focus on completing mechanical fabrication, integrating the vision, control, and gripper subsystems, and conducting systematic experimental evaluations. Performance metrics such as insertion success rate, cycle time, repeatability, and misalignment recovery effectiveness will be used to assess system performance and compare it with manual assembly processes. Overall, this project establishes a strong foundation for demonstrating how mechanical compliance and intelligent system integration can enhance flexibility, robustness, and efficiency in automated wire and cable assembly applications.

Bibliography

[1] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 4th ed. Boston, MA, USA: Pearson, 2018.

[2] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, *Robotics: Modelling, Planning and Control*. London, U.K.: Springer, 2009.

[3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[4] M. T. Mason, "Compliance and force control for computer controlled manipulators," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-11, no. 6, pp. 418–432, Jun. 1981.

[5] N. Hogan, "Impedance control: An approach to manipulation," *Journal of Dynamic Systems, Measurement, and Control*, vol. 107, no. 1, pp. 1–24, Mar. 1985.

[6] S. Haddadin, A. Albu-Schäffer, and G. Hirzinger, "Safety evaluation of physical human–robot interaction via crash-testing," *Robotics: Science and Systems*, pp. 217–224, 2007.

[7] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed. Cambridge, MA, USA: MIT Press, 2011.

[8] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2004.

[9] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. New York, NY, USA: Pearson, 2018.

[10] A. Bicchi and V. Kumar, "Robotic grasping and contact: A review," *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 348–353, 2000.

[11] K. B. Shimoga, "Robot grasp synthesis algorithms: A survey," *International Journal of Robotics Research*, vol. 15, no. 3, pp. 230–266, Jun. 1996.

[12] ABB Robotics, *ABB RobotStudio Operating Manual*, ABB Group, Zurich, Switzerland, 2022.

[13] F. J. Ruiz-Ruiz, "Compliant gripper with force estimation for physical human–robot interaction," *Robotics and Mechatronics Group*, Universidad de Málaga, Spain, Dec. 2022.